

Chronic Administration of Belimumab, a BLyS Antagonist, Decreases Tissue and Peripheral Blood B-Lymphocyte Populations in Cynomolgus Monkeys: Pharmacokinetic, Pharmacodynamic, and Toxicologic Effects

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Received December 19, 2005; accepted February 23, 2006

The tolerability, pharmacodynamic effects, and pharmacokinetics of belimumab (LymphoStat-B) were evaluated in cynomolgus monkeys. Belimumab is a fully human IgG1 λ antibody directed against B-lymphocyte stimulator (BLyS) protein. BLyS is a TNF family member that supports B-lymphocyte maturation and survival and has been implicated in the pathogenesis of autoimmune diseases and B-lymphocyte malignancies. Belimumab was developed to antagonize BLyS activity in autoimmune diseases and B-lymphocyte malignancies, where undesirable effects of B-lymphocyte activity may cause or contribute to disease. Pharmacodynamic effects of belimumab were monitored by immunophenotyping of peripheral blood. Pathology end points, including tissue immunophenotyping, are described after 13 and 26 weeks of treatment and after a 34-week treatment-free (recovery) period. Belimumab was safe and well tolerated in repeat-dose toxicology studies at 5–50 mg/kg for up to 26 weeks. Monkeys exposed to belimumab had significant decreases in peripheral blood B lymphocytes by 13 weeks of exposure, continuing into the recovery period, despite total lymphocyte counts similar to the controls. There were concomitant decreases in spleen and lymph node B-lymphocyte representation after 13 or 26 weeks of treatment with belimumab. Microscopically, monkeys treated with belimumab for 13 or 26 weeks had decreases in the number and size of lymphoid follicles in the white pulp of the spleen. All findings were generally reversible within a 34-week recovery period. These data confirm the specific pharmacologic activity of belimumab in reducing B lymphocytes in the cynomolgus monkey. The favorable safety profile and lack of treatment-related infections also support continued clinical development of belimumab.

Key Words: agents—pharmaceuticals; immunotoxicology—autoimmune; safety evaluation; safety evaluation—toxicity; chronic.

Belimumab (LymphoStat-B; Human Genome Sciences, Rockville, MD) is a recombinant fully human IgG1 λ monoclonal antibody that targets the B-lymphocyte stimulator (BLyS) protein (Baker *et al.*, 2003). BLyS is a member of the tumor necrosis factor (TNF) family and is also referred to as TNF homolog that activates apoptosis, NF- κ B, and JNK; TNF- and ApoL-related leukocyte-expressed ligand 1; or B-cell activating factor belonging to the TNF family (BAFF) (Moore *et al.*, 1999; Mukhopadhyay *et al.*, 1999; Schneider *et al.*, 1999; Shu *et al.*, 1999). BLyS is produced by myeloid cells as a type II transmembrane protein, which is cleaved to form the soluble, biologically active form (Nardelli *et al.*, 2001, and reviewed in Stohl, 2005). Recombinant soluble BLyS can enhance B lymphocyte activation and prolong B lymphocyte survival (Do *et al.*, 2000; Gross *et al.*, 2001; Thompson *et al.*, 2000; and reviewed in Mackay and Browning, 2002), both of which are thought to contribute to the phenotype of autoimmune diseases like systemic lupus erythematosus (SLE).

There are at least three receptors for BLyS: B-cell maturation antigen (BCMA), transmembrane activator and calcium modulator and cyclophilin ligand interactor (TACI), and BLyS receptor 3 (BR3), also called BAFF receptor; all three are predominantly expressed on the surface of B lymphocytes (Gross *et al.*, 2000; Marsters *et al.*, 2000; Thompson *et al.*, 2001; Wu *et al.*, 2000; Yu *et al.*, 2000). Of these, BR3 is considered responsible for many of the known activities of BLyS (Gross *et al.*, 2001; Shulga-Morskaya *et al.*, 2004; Thompson *et al.*, 2001).

Overproduction of BLyS in transgenic mice is associated with lymphoid hyperplasia and features of autoimmunity (Khare *et al.*, 2000; Mackay *et al.*, 1999). Evidence of increased circulating BLyS levels in patients with autoimmune diseases has also been reported (Stohl *et al.*, 2003). Importantly, BLyS appears to have limited or no effects on B-lymphocyte precursors (pre-B cells in the bone marrow), which express CD20 but do not express any of the known BLyS receptors. Thus, BLyS antagonism represents a desirable mechanism for a reversible

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targeted therapeutic to attenuate mature B-lymphocyte activity in autoimmune diseases. In addition, it should be noted that although human IgG1 antibodies have the potential to induce antibody-dependent cell-mediated cytotoxicity (ADCC), belimumab specifically recognizes the soluble, biologically active form of BLyS (Baker *et al.*, 2003). Therefore, ADCC should not be elicited by this antibody/antigen interaction.

Although recombinant human BLyS has pharmacologic activity in mice (Parry *et al.*, 2001), belimumab does not cross-react with the mouse BLyS homolog. Limited studies of belimumab in rodents were conducted and demonstrated that belimumab could block the B-lymphocyte expansion observed with the administration of recombinant human BLyS (Baker *et al.*, 2003).

Belimumab is currently in clinical development for use in the treatment of autoimmune diseases such as SLE and rheumatoid arthritis. In a phase 1 safety study, belimumab was well tolerated and demonstrated biological activity after one or two infusions at 1, 4, 10, or 20 mg/kg/dose in patients with SLE (Furie *et al.*, 2003). The studies described here were conducted to better predict and understand chronic antibody-mediated removal of BLyS from the circulation.

The cynomolgus monkey was chosen as a relevant species for the nonclinical evaluation of belimumab. There is a high degree of homology between cynomolgus BLyS and human BLyS (96.4% identity at the amino acid level) and similar binding affinities of belimumab to human or cynomolgus BLyS in ELISA-based assays (Baker and Wu, Human Genome Sciences [HGS], data not shown). The B-lymphocyte (CD20+) fraction of isolated cynomolgus peripheral blood mononuclear cells expresses BLyS receptors at levels comparable to those on human B lymphocytes (HGS, data not shown). Finally, activity of belimumab in cynomolgus monkeys was demonstrated in a previously reported study, with decreased spleen and lymph node B-lymphocyte representation after four weekly iv administrations (Baker *et al.*, 2003).

We hypothesized that antagonism of BLyS activity by belimumab would specifically decrease B-lymphocyte populations and that these effects would be most apparent after chronic administration, consistent with a decrease in B-lymphocyte proliferation and/or survival. To evaluate this hypothesis, multiple *in vivo* studies of belimumab were performed in monkeys. The pharmacokinetic profile of belimumab was determined in cynomolgus monkeys after a single-dose administration and was used to model predicted exposures to establish an appropriate schedule for continuous exposure with multiple-dose administrations in the monkey. In multiple-dose studies, including up to 26 weeks of dosing, pharmacokinetic monitoring was used to confirm expected exposures.

The safety, tolerability, and pharmacodynamic effects of belimumab after iv administration to cynomolgus monkeys are reported for up to 26 weeks of dosing. The majority of results included are from a chronic (26-week) administration study described here in detail. Limited relevant information from other iv administration studies of belimumab in cynomolgus

monkeys, including pharmacokinetics studies and additional data from a previously described 4-week toxicology study, is included as well.

MATERIALS AND METHODS

Belimumab. Belimumab is a fully human IgG1 λ monoclonal antibody that specifically binds to and antagonizes the activity of human BLyS. The generation and characterization of belimumab have been described previously (Baker *et al.*, 2003). Briefly, a human phage display library was screened for antibodies against human BLyS, and belimumab was chosen after affinity maturation of an antibody from the initial screening process. Antibody was purified from production cell supernatants using a series of chromatography steps. The final belimumab product used in these studies was lyophilized and stored at 2–8°C prior to reconstitution and use. The vehicle for belimumab (1.9% glycine, 0.5% sucrose, 10mM sodium citrate, and 0.01% Tween 80, pH 6.5 $\pm$ 0.4) was used both as a diluent and as the control material for all studies.

Pharmacokinetics of belimumab. The pharmacokinetics of belimumab after iv administration have been evaluated in single- and multiple-dose studies in cynomolgus monkeys. The pharmacokinetic analysis is most complete for the single-dose and 4-week study. The pharmacokinetics determined from the single-dose and 4-week studies are included here to support the interpretation of the more limited pharmacokinetics from the 26-week study.

Groups of four cynomolgus monkeys were injected with single iv doses of 5 or 50 mg/kg of belimumab, administered in 2.5 ml/kg. In a separate study, a dose of 150 mg/kg (7.5 ml/kg) was evaluated in three female monkeys. Samples for serum concentration analysis were obtained at 13 points for up to 64 days postdosing. Belimumab concentrations in serum samples were determined with a sandwich-type ELISA that utilized BLyS for capture and goat HRP-conjugated anti-human IgG for detection (note: a slightly different assay, utilizing a mouse monoclonal secondary antibody reagent, was employed for the 150-mg/kg single-dose pharmacokinetic study).

The dose range of 5–50 mg/kg/dose was employed in two repeat-dose toxicology studies of 4 and 26 weeks, with periodic serum sampling to confirm exposure. In the 26-week toxicology study, serum samples for drug concentration analysis were obtained every 2 weeks immediately prior to each dose administration and throughout the recovery period of the study.

In the single-dose study, serum concentration data for each individual monkey were fit to a two- or three-compartment model with first-order elimination from the central compartment using the software package WinNonlin (Pharsight Corp., Mountain View, CA). In the multiple-dose studies, non-compartmental analysis with linear up/log down trapezoidal analysis and uniform weighting of the log-transformed data were used to determine half-lives following the final dose of drug.

Toxicology studies. A 4-week repeat-dose, iv administration toxicology study of belimumab administered weekly at 0, 5, 15, or 50 mg/kg/dose (2.5-ml/kg dose volume), conducted according to Good Laboratory Practice (GLP) standards, has been previously described (Baker *et al.*, 2003). Additional limited data (not previously reported) from that study, including pharmacokinetics analysis, immunophenotyping of peripheral blood mononuclear cells, and macroscopic and microscopic observations, are described in this report.

A 26-week iv administration toxicology study was conducted according to GLP standards at Charles River Laboratories Preclinical Services, Nevada (Nevada); all study-specific procedures were conducted by Nevada, HGS, or, as indicated, an HGS- or Nevada-sponsored subcontractor. Animal care was in full compliance with the regulations outlined in the USDA Animal Welfare Act (9 CFR, Parts 1, 2, and 3) and the conditions specified in the Guide for the Care and Use of Laboratory Animals (ILAR publication, 1996, National Academy Press).

Experimentally naive cynomolgus monkeys (*Macaca fascicularis*) underwent a comprehensive health evaluation and screening, which included

demonstration of seronegativity and/or PCR-negativity for simian retroviruses. There were 60 monkeys selected for the 26-week study. Male ($n = 30$; b.wt. 2.0–4.7 kg) and female ($n = 30$; b.wt. 1.7–3.5 kg) monkeys were each assigned to one of four treatment groups by a stratified randomization scheme designed to achieve similar group mean body weights and with equal numbers of male and female monkeys in each group.

There were six male and six female monkeys in the control group and eight male and eight female monkeys in each of the three belimumab treatment groups. Monkeys were acclimated to nonhuman primate chairs and were not sedated for dose administration.

Dose levels were 0 mg/kg (Group 1), 5 mg/kg (Group 2), 15 mg/kg (Group 3), or 50 mg/kg (Group 4) for each dose administration. Belimumab, or diluent control, was iv-administered in a dose volume of 2.5 ml/kg to all monkeys via a peripheral vein once every 2 weeks for 13 or 26 weeks (7 or 13 times) in the chronic administration study. The every 2-week schedule was selected based on available pharmacokinetics data of belimumab in monkeys and was intended to be as frequent as or more frequent than the clinical dose schedule, with a target of continuous exposure during the treatment period.

Monkeys were observed, including estimated food consumption, at least twice daily starting 7 days prior to the first dose administration and continuing through the day of scheduled necropsy. Body weights were determined prior to dose administration and once weekly for the duration of the study. Electrocardiograms and ophthalmic examinations were performed under light sedation prior to the first dose administration (as part of the health screening), and during weeks 13, 26, and 39 in the chronic study. Electrocardiogram recordings were evaluated by ANILAB (Englishtown, NJ).

Flow cytometry. Blood samples (approximately 1 ml) for the evaluation of peripheral blood mononuclear cell populations were collected into heparin-containing tubes from all available monkeys periodically from each study, including samples from before, during, and after each treatment period. Tissue leukocytes for the evaluation of spleen and lymph node mononuclear cell populations were extracted from spleen and mesenteric lymph node at the time of each necropsy. Samples for flow cytometry were evaluated using a Coulter Epics XL-MCL instrument (Beckman-Coulter, Miami, FL). For both peripheral blood and tissue samples, the following markers that have been qualified for use in cynomolgus monkey specimens were used to distinguish the immunophenotype: CD2 for total lymphocytes, CD20 for total B lymphocytes, CD20 and CD21 for mature B lymphocytes, CD3 for total T lymphocytes, CD3 and CD4 for helper T lymphocytes, CD3 and CD8 for suppressor/cytotoxic T lymphocytes, and CD14 (CD3[−]) for monocytes. The PE-labeled anti-CD2 antibody was obtained from Beckman-Coulter. FITC-labeled anti-CD3 and anti-CD20 antibodies, as well as PE-labeled anti-CD4, anti-CD8, anti-CD14, and anti-CD21 antibodies, were obtained from BD PharMingen (San Diego, CA).

Clinical pathology. Blood samples for the evaluation of serum chemistry, hematology, and coagulation parameters were collected periodically from all available monkeys. Approximate blood volumes were 2 ml for serum chemistry, 1 ml for hematology, and 1.8 ml for coagulation. Monkeys were fasted overnight prior to blood collection for clinical pathology.

For serum chemistry samples, sera were analyzed on a Beckman Synchron CX7 automated chemistry analyzer (Beckman-Coulter Instruments, Palo Alto, CA). The following parameters were analyzed: sodium, potassium, chloride, carbon dioxide, total bilirubin, alkaline phosphatase, lactate dehydrogenase, aspartate aminotransferase, alanine aminotransferase, gamma-glutamyl transferase, calcium, phosphorus, urea nitrogen, creatinine, total protein, albumin, globulin, albumin/globulin ratio, glucose, cholesterol, and triglycerides.

Standard hematology evaluation of whole blood collected into tubes containing EDTA anticoagulant included: red blood cell counts, white blood cells (total and differential), hemoglobin concentration, hematocrit, reticulocyte counts, mean cell hemoglobin, mean corpuscular volume, mean corpuscular hemoglobin concentration, platelet counts, and blood cell morphology. Samples were analyzed on an Abbott Cell-Dyn 3500 multiparameter automated hematology analyzer (Abbott Labs, Pomezia, Italy). In addition, total lymphocyte counts were periodically determined for use in combination with flow cytometry data.

For assessment of coagulation parameters, plasma from whole blood collected into tubes containing sodium citrate anticoagulant was analyzed for activated partial thromboplastin time, prothrombin time, and fibrinogen on a Sigma AMAX CS 190 (Sigma Diagnostics, St. Louis, MO).

Urinalysis was performed on samples obtained from the bladder at necropsy. The following parameters were determined: color/character (observation); specific gravity (refractometer); pH, leukocyte esterase, nitrite, urobilinogen, protein, glucose, ketones, bilirubin, and occult blood (dipstick); and casts, leukocytes, erythrocytes, epithelial cells, mucus, crystals, bacteria, yeast, and amorphous sediment (microscopic evaluation of sediment).

Immunoglobulin subclasses. Serum samples were obtained from each available monkey at weeks 1 (predose), 6, 13, 22, 26, 34, 39, 52, and 60 and were evaluated for cynomolgus monkey IgG, IgM, IgA, and IgE. Concentration determination for immunoglobulin subclasses was performed by AniLytics (Gaithersburg, MD) using an ELISA-based method. Assessment of immunoglobulin subclasses was performed on all available samples collected as of the time of each necropsy (weeks 13, 26, and 60). Results were reported for each sample in units of milligrams per deciliter for IgM, IgA, and IgG or as nanograms per milliliter for IgE.

Anti-belimumab antibodies. The presence of belimumab-specific antibodies in serum was determined prior to first dose administration and periodically during the chronic study. This analysis utilized a sandwich-type ELISA, with detection using human-specific anti-IgA, anti-IgM, anti-IgG-Fc, and anti-kappa light-chain antibodies. This evaluation was performed after 13 weeks of treatment, after 26 weeks of treatment, and at weeks 52 and 60 following a 26- or 34-week treatment-free period. The presence of monkey antibodies to belimumab was defined as an increase in A_{450} of at least twofold in the sera obtained posttreatment compared to the A_{450} obtained from the predose sera from the same monkey.

Anatomic pathology. Necropsies were scheduled after 13 weeks of treatment ($n = 22$), 26 weeks of treatment ($n = 22$), or 26 weeks of treatment followed by a 34-week treatment-free (recovery) period ($n = 16$). During the necropsies conducted at weeks 13 and 26, three male and three female monkeys from each group treated with belimumab were evaluated at each time point and two male and two female monkeys from each group were evaluated after the recovery period. At each of the three scheduled necropsies, two male and two female monkeys from the control group were evaluated.

A complete gross necropsy was conducted on each monkey. Organ weights were determined for a standard set of tissues and analyzed for treatment effects on the organ weight and as a ratio of the organ weight to body weight and to brain weight. A standard list of tissues was collected, preserved, and examined histologically by a pathologist (P.L.); selected tissues were peer reviewed by a second pathologist (W.G.H.).

Statistics. In general, group means and standard deviation values were calculated for numerical data, including body weights, circulating lymphocyte and monocyte subpopulations, clinical pathology parameters, organ weights, and lymphoid tissue lymphocyte subset and monocyte populations. Further statistical analyses were performed with SAS System version 8.1. Significant intergroup differences were evaluated by the use of an ANOVA. If the parametric ANOVA was significant at $p \leq 0.05$, Dunnett's test was used to identify statistically significant differences between the control group and each belimumab-treated group at the $p \leq 0.05$ level of significance.

For the immunoglobulin subclass treatment period data (weeks 1–26 of the chronic toxicology study), a repeated measures model was used to analyze the Ig level across time points. In addition to treatment and time, the model was adjusted for gender and predose Ig level. For IgG, for example, the model is

$$\text{IgG} = \text{Predose IgG} + \text{Gender} + \text{Treatment} + \text{Week} + \text{Treatment} \times \text{Week}.$$

This approach also accounted for the correlation between repeated measures within each subject by employing a compound symmetric covariance structure. The MIXED procedure in SAS version 8 was used to estimate the model

parameters. If overall treatment differences were identified, *post hoc* tests were performed comparing each belimumab treatment group to control.

For the recovery period data, there were only four monkeys in each treatment group at each time point, which is insufficient to estimate overall effects of time and treatment while adjusting for the correlation of observations within each subject. Therefore, a one-way ANOVA at each time point was used to compare the effect of treatment on total Ig as well as each of the Ig subclasses. If the *F*-test indicated some evidence of treatment differences, *post hoc* tests were performed, comparing each belimumab treatment group to control.

RESULTS

Serum Belimumab Concentrations

Mean serum belimumab concentration time profiles following a single dose are presented in Figure 1 and after multiple-dose administrations in Figures 2 and 3. Single-dose pharmacokinetic parameters are listed in Table 1 for each dose level evaluated and in Table 2 for repeated dose administrations. Following iv administration, the serum belimumab concentration declined in a multiexponential manner with a mean terminal half-life of 11–14 days. In the single-dose study, clearance ranged between an average of 5.6 and 6.8 ml/day/kg and the volume of distribution at steady state is between 85 and 108 ml/kg. As illustrated in Table 2, the pharmacokinetics of belimumab following four weekly iv injections concur with the pharmacokinetic parameters obtained in the single-dose study, indicating that the pharmacokinetics of belimumab do not change after multiple doses.

Complete pharmacokinetic parameters were not determined in the 26-week study, but exposure was similar to that predicted by the single-dose pharmacokinetics. In addition, the elimination half-lives determined for multiple-dose administrations in monkeys were similar to the single-dose terminal half-life, with values ranging between 9 and 16 days. Exposure profiles

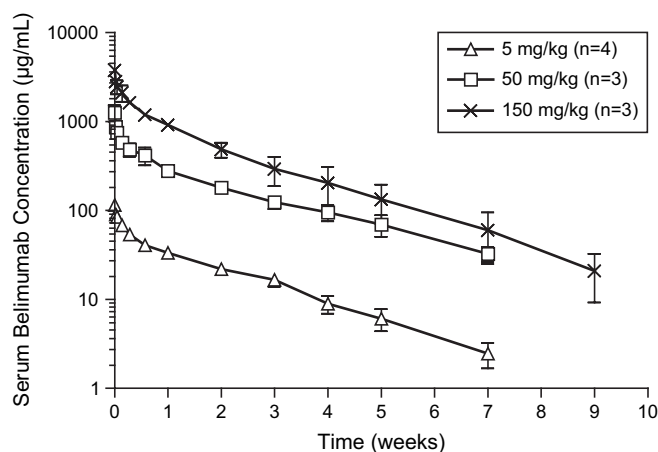


FIG. 1. Mean serum belimumab concentrations $\pm$ SD following a single iv dose administration to cynomolgus monkeys at 5 mg/kg ($n = 4$), 50 mg/kg ($n = 3$), or 150 mg/kg ($n = 3$).

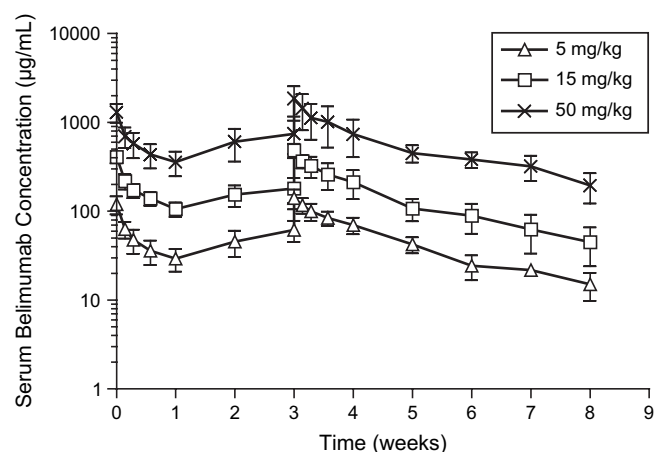


FIG. 2. Mean serum concentrations of belimumab ($\pm$ SD) are presented by treatment group. The treatment period included days 1–29, with dose administrations every 7 days. $n = 10$ monkeys per group through day 29, $n = 4$ monkeys per group through day 56.

determined from serum samples collected every 2 weeks, immediately prior to each dose administration, are illustrated in Figure 8 for the 26-week study. It should be noted that in the 26-week study, two monkeys developed anti-belimumab antibodies and were excluded from the analysis.

General Observations

Belimumab was consistently well tolerated when administered iv to cynomolgus monkeys in the 26-week study described, as well as in prior toxicology and pharmacokinetic studies. With administration for up to 26 weeks, there were no cage side observations (including food consumption), changes

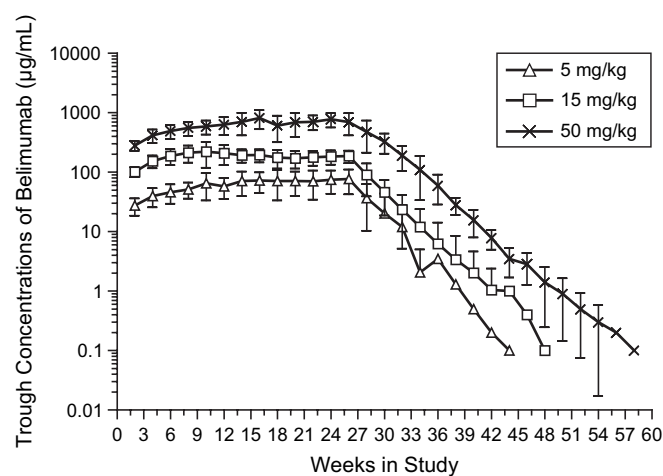


FIG. 3. Mean trough serum concentrations of belimumab ($\pm$ SD) are presented by treatment group. The treatment period included weeks 1–26, with dose administrations every 2 weeks. $n = 16$ monkeys per group through week 13, $n = 10$ monkeys per group from weeks 14 through 26, and $n = 4$ monkeys per group from weeks 27 through 60.

TABLE 1
Single-Dose Pharmacokinetics of iv Belimumab in
Cynomolgus Monkeys

Parameters	5 mg/kg (n = 4)	50 mg/kg (n = 3) ^a	150 mg/kg (n = 3)
C _{max} (μg/ml)	111 ± 2	1230 ± 180	3778 ± 326
AUC _{0-∞} (day μg/ml)	901 ± 40	8970 ± 320	22,693 ± 3990
V ₁ (ml/kg)	45.0 ± 0.7	42.4 ± 5.9	39.7 ± 3.3
V _{ss} (ml/kg)	85.0 ± 2.1	108 ± 14	89.8 ± 11.4
CL (ml/day/kg)	5.6 ± 0.2	5.6 ± 0.2	6.8 ± 1.3
t _{1/2,α} (day)	0.52 ± 0.13	0.44 ± 0.20	0.41 ± 0.55
t _{1/2,term} (day)	11.2 ± 0.7	14.0 ± 1.8	11.5 ± 1.7
MRT (day)	15.4 ± 1.0	19.1 ± 2.1	13.7 ± 3.5

Note. C_{max}, maximum serum drug concentration; AUC_{0-∞}, area under the serum drug concentration-time curve from time 0 to infinite time; V₁, volume of distribution for the central compartment; V_{ss}, volume of distribution at steady state; CL, clearance; MRT, mean residence time; t_{1/2,α}, elimination half-life for the distribution phase; t_{1/2,term}, elimination half-life for the terminal phase.

^aOne animal developed anti-belimumab antibodies and was excluded from the mean calculation.

in body weight, electrocardiographic findings, or ophthalmic findings that were attributed to belimumab administration. Most cage side observations were considered typical of background observations in monkeys undergoing study procedures, were distributed similarly across all treatment groups including controls, and occurred both in multiple studies and, where applicable, in different rooms for the same study. During the chronic administration study, sneezing and red nasal discharge were noted in 12 monkeys, including monkeys from both con-

TABLE 2
Pharmacokinetics of Belimumab after 4 Weekly iv Doses in
Male and Female Cynomolgus Monkeys

Parameters ^a	5 mg/kg (n = 4)	15 mg/kg (n = 4)	50 mg/kg (n = 4)
C _{max} (μg/ml)	102 ± 14	313 ± 38	1064 ± 190
AUC _{0-∞} (day μg/ml)	717 ± 129	2365 ± 501	9286 ± 1809
V ₁ (ml/kg)	49.7 ± 6.8	48.4 ± 6.2	48.0 ± 7.8
V _{ss} (ml/kg)	126 ± 18	108 ± 10	106 ± 18
CL (ml/day/kg)	7.2 ± 1.5	6.5 ± 1.2	5.5 ± 1.1
t _{1/2,α} (day)	0.71 ± 0.26	1.29 ± 0.61	0.77 ± 0.16
t _{1/2,term} (day)	13.5 ± 1.8	14.0 ± 3.1	14.6 ± 2.2
MRT (day)	17.8 ± 2.5	17.0 ± 4.2	19.4 ± 3.5

Note. C_{max}, maximum serum drug concentration; AUC_{0-∞}, area under the serum drug concentration-time curve from time 0 to infinite time; V₁, volume of distribution for the central compartment; V_{ss}, volume of distribution at steady state; CL, clearance; MRT, mean residence time; t_{1/2,α}, elimination half-life for the distribution phase; t_{1/2,term}, elimination half-life for the terminal phase.

^aPK parameters were estimated using serum concentration data from the two male and two female monkeys in each group that were followed through the recovery period.

trol and belimumab-treatment groups, between days 123 and 133. Four additional (different) monkeys had similar symptoms for 2–9 days between days 231 and 265, and all the affected monkeys had been housed in the same room. A bacterial infection (*Branhamella catarrhalis*) was considered the likely cause of this finding, and all affected monkeys responded to treatment with antibiotic (penicillin). Treatment with antibiotics was limited to monkeys with overt clinical signs. Due to the incidence in both control and treated monkeys, restriction to one room, and response to antibiotic treatment, this finding was not directly attributed to potential immunomodulatory properties of belimumab. However, a distinction between no effect or mild effect of belimumab on susceptibility to this infection was not possible with the small numbers of monkeys affected during each of the two “outbreaks” in this room.

Peripheral Blood Mononuclear Cells

In the previously conducted 4-week toxicology study, weekly administration of belimumab at 5, 15, or 50 mg/kg/dose for 4 weeks resulted in no apparent decreases in circulating CD20+ lymphocytes during the treatment period, and only a trend toward decreased CD20+ cells at the end of a 4-week treatment-free recovery period (Fig. 4A). This suggests a requirement for treatment for longer than 4 weeks to induce the reduction in circulating peripheral B lymphocytes by belimumab in healthy cynomolgus monkeys.

In contrast, treatment with belimumab at 5, 15, or 50 mg/kg/dose every other week for 13 or 26 weeks resulted in the expected pharmacologic effect of decreasing CD20+ B lymphocytes in the peripheral blood (Fig. 4B). There was an even greater effect on a mature subset of circulating B lymphocytes (CD20+ and CD21+; Fig. 4C). There were no concurrent changes in total lymphocyte numbers that were attributed to belimumab (Fig. 4D).

The effects of belimumab appeared to be dose dependent after 13 weeks of treatment and maximal in all dose groups by week 26, with mean CD20+ lymphocytes of 41, 42, and 35% of the baseline for the 5-, 15-, and 50-mg/kg dose groups, respectively. The corresponding value for the control group was 99% of the baseline. Despite cessation of treatment at week 26, the decrease in circulating peripheral blood B lymphocytes persisted in belimumab-treated monkeys through at least week 39 and in some monkeys longer. The magnitude of decrease was dose independent within groups treated for 26 weeks with belimumab. In general, B-lymphocyte populations had returned to normal levels by week 60, with some monkeys still slightly below baseline at the week 60 sampling time. However, definitive interpretation of reversibility of this effect was hampered by the small numbers of monkeys in the recovery groups and the variability of the data inherent in cynomolgus monkeys.

All other mononuclear cell subsets evaluated, including T-lymphocyte subsets and monocytes, were similar between

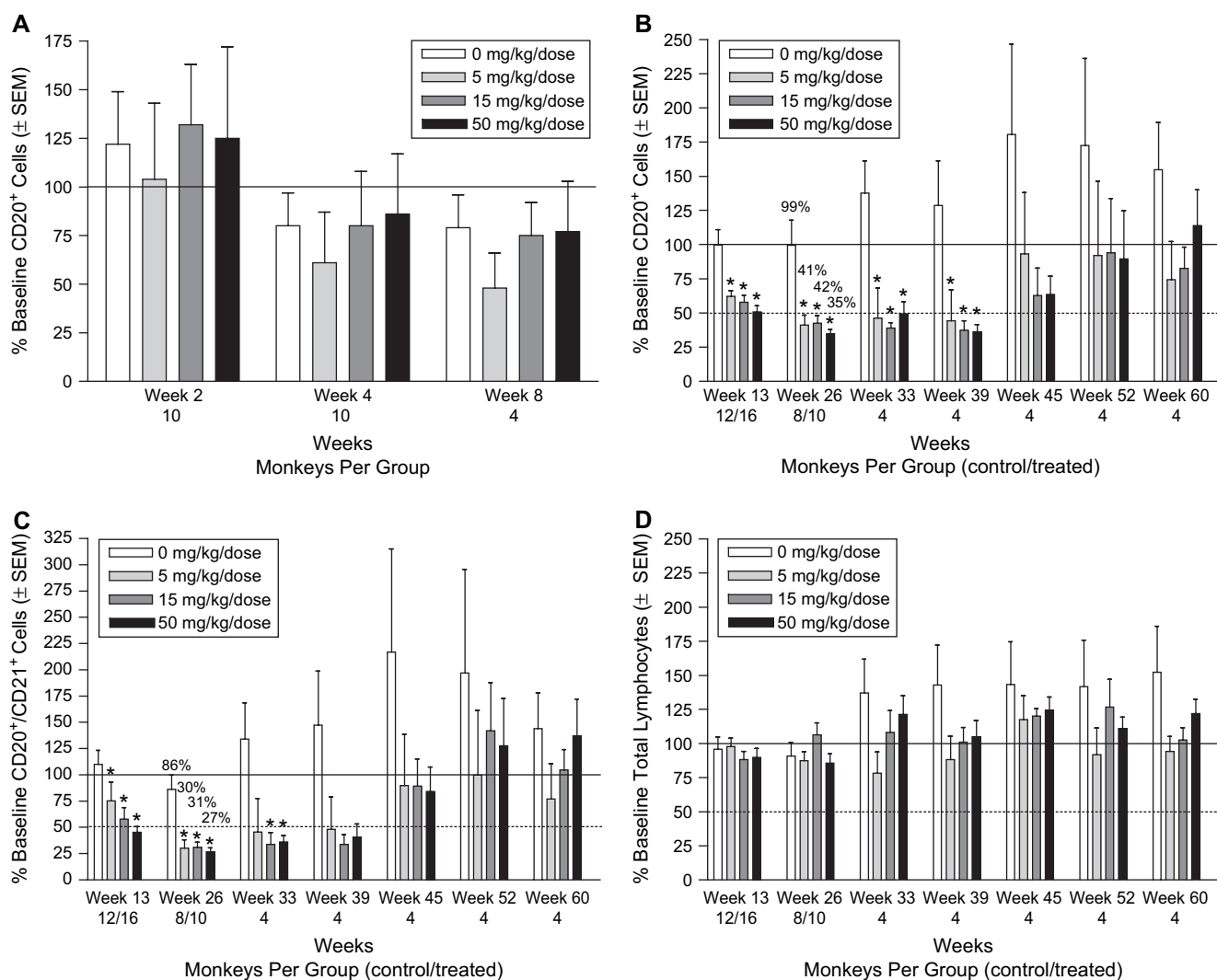


FIG. 4. Peripheral blood lymphocyte representation as a percentage of baseline levels according to belimumab dose level. Bars indicate mean percentage of baseline $\pm$ SEM; the N for each time point is labeled for vehicle and belimumab-treated groups. For each study, baseline was the average of three measurements taken prior to initiation of dosing. (A) Peripheral blood CD20⁺ cells as a percentage of baseline levels ($\pm$ SEM) after 4 weeks of treatment ($n = 10$ per dose level), and after 4 weeks of treatment followed by a 4-week treatment-free recovery period ($n = 4$ per dose level). (B–D) Belimumab, or vehicle control, was administered iv every other week for 13 or 26 weeks, and peripheral blood lymphocytes were assessed by flow cytometric methods periodically over the treatment period and a subsequent 34-week treatment-free recovery period. Asterisks indicate significant decreases in B lymphocytes as compared to baseline ($p < 0.05$). The actual percentage of baseline is indicated for each treatment group at week 26. Assessment included CD20⁺ B lymphocytes (B), CD20⁺/CD21⁺ mature B lymphocytes (C), and total lymphocytes (D).

belimumab-treated and control monkeys at each time point and comparable to established baseline values for each parameter (data not shown).

Tissue Mononuclear Cells

The early effects of 4 weeks of treatment with belimumab on lymphoid tissue B lymphocytes in cynomolgus monkeys have been previously reported (Baker *et al.*, 2003). In that study, statistically significant decreases in the relative percentage of both CD20⁺ and CD20/CD21 dual-positive mononuclear cells were observed in the mesenteric lymph node and

spleen from all three belimumab-treated groups (as compared to vehicle controls), but not until the end of the 4-week recovery period. There were no statistically significant differences between belimumab-treated groups for lymphoid tissues collected at the end of the 4-week treatment period.

In the chronic study reported here, there were decreases in the relative percentage of B lymphocytes in the spleen and mesenteric lymph node of monkeys in all belimumab-treatment groups compared to vehicle controls when evaluated at weeks 13 and 26. There was no apparent dose dependency of effect on tissue B lymphocytes in monkeys treated with belimumab at 5, 15, or 50 mg/kg during the treatment period. The mean

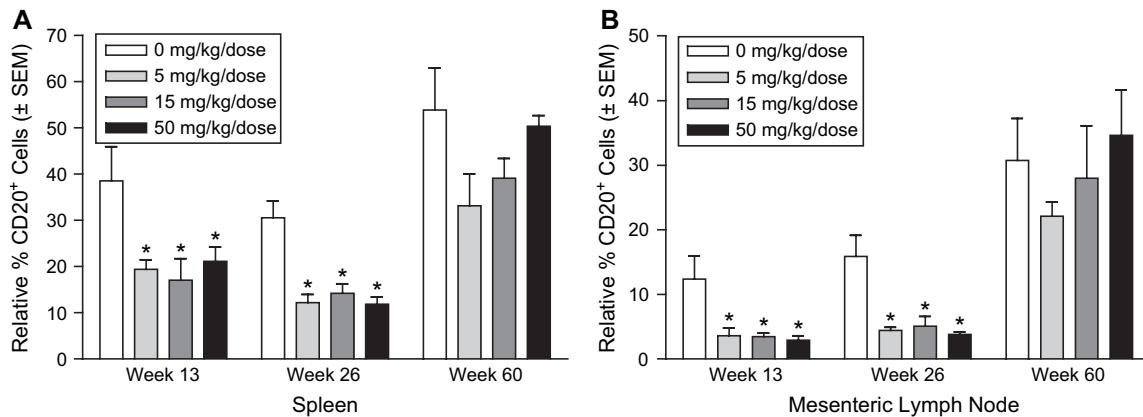


FIG. 5. CD20+ B lymphocytes in tissue measured by flow cytometry at necropsy is presented as the mean percentage of cells isolated from spleen (A) or mesenteric lymph node (B). Bars indicate the mean $\pm$ SEM for each dose level. $n = 6$ for belimumab-treated monkeys at 13 and 26 weeks; $n = 4$ for vehicle control monkeys at 13 and 26 weeks and for all groups at 60 weeks. Asterisks indicate significant decreases in B lymphocytes in belimumab-treated monkeys as compared to vehicle controls ($p < 0.05$).

values for weeks 13, 26, and 60 are shown in Figures 5 and 6 for the percentages of CD20+ and CD20+/CD21+ lymphocytes, respectively. At week 13, mean values of percent CD20+ and CD20+/CD21+ B lymphocytes in all groups of monkeys treated with belimumab were on the order of approximately 50 and 30%, respectively, of control values in the spleen. In the mesenteric lymph node, mean values of percent CD20+ cells in all groups of belimumab-treated monkeys at week 13 were approximately 30% of control values, while CD20+/CD21+ reductions were of greater magnitude but highly variable. At week 26, there were substantial reductions in CD20+ and CD20+/CD21+ B lymphocytes in both the spleen (25–50% of mean control) and mesenteric lymph node (10–30% of mean control) of all groups of belimumab-treated monkeys. The

overall reductions in tissue B lymphocytes corresponded to the decreased number of these cell subtypes measured in the peripheral blood.

At week 60 (after a 34-week recovery period), the percentage of CD20+ and CD20+/CD21+ B lymphocytes in the spleen and mesenteric lymph node, most often in low-dose (5 mg/kg) monkeys, remained slightly below the mean percentages of these parameters from corresponding control monkeys. This pattern also corresponded to the pattern noted in peripheral blood B lymphocytes at weeks 33 and 60. These apparent reductions present in belimumab-treated monkeys at week 60 were considered within the limits of normal variability. All other mononuclear cell subsets were similar across groups, including the vehicle control group, for weeks 13, 26, and 60.

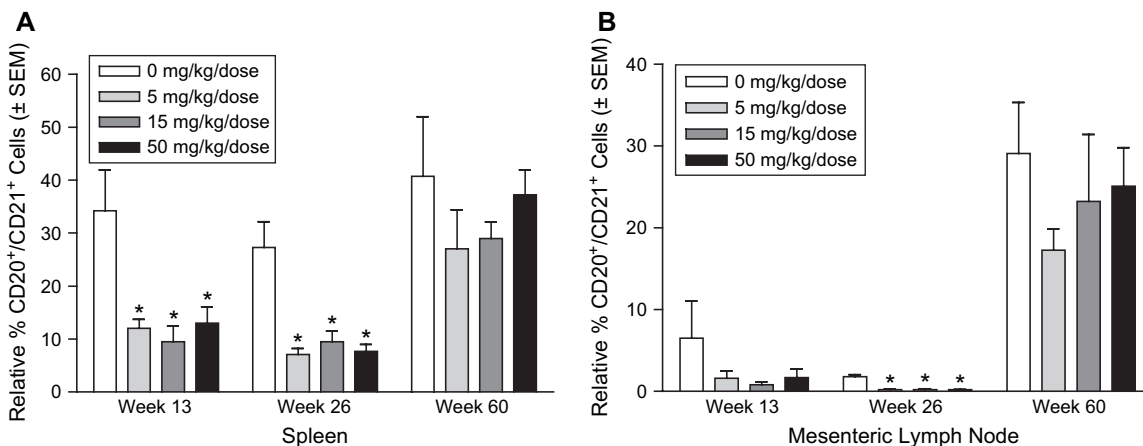


FIG. 6. CD20+/CD21+ B lymphocytes (mature B lymphocytes) in tissue measured by flow cytometry at necropsy is presented as the mean percentage of cells isolated from spleen (A) or mesenteric lymph node (B). Bars indicate the mean $\pm$ SEM for each dose level. $n = 6$ for belimumab-treated monkeys at 13 and 26 weeks; $n = 4$ for vehicle control monkeys at 13 and 26 weeks and for all groups at 60 weeks. Asterisks indicate significant decreases in B lymphocytes in belimumab-treated monkeys as compared to vehicle controls ($p < 0.05$).

Gross Organ Weights

In general, there were no clear effects of belimumab on organ weights in the prior 4-week toxicology study (data not shown) or after 13 or 26 weeks of exposure in the chronic toxicology study. However, after 26 weeks of treatment with belimumab, there was a trend toward reduction in splenic weight in some monkeys from all belimumab dose groups, as compared to vehicle controls. When correlated to the severity of decreases in lymphoid follicle size/number described below, this observation is more readily appreciated. Figure 7A illustrates the spleen weight (g) according to dose group, and Figure 7B illustrates the same data represented according to the severity of lymphoid follicular depletion observed microscopically. As represented in Figure 7B, a roughly 25–50% decrease in spleen weight was identified after 26 weeks of treatment when spleen weights from monkeys with microscopic evidence of decreased lymphoid follicle size and number were compared to spleen weights from vehicle control monkeys. In contrast, there was no appreciable difference in spleen weights from belimumab-treated monkeys with microscopically normal appearing splenic follicles as compared to spleen weights of vehicle controls.

Macroscopic Examination

There were no macroscopic findings that could be clearly associated with belimumab in the 4-week or chronic studies. In the 4-week toxicology study of belimumab, a single monkey in the 50-mg/kg dose group had multifocal splenic abscesses that were identified at necropsy and considered to be possibly related to the potentially immunomodulatory effects of belimumab treatment. However, there were no gross lesions of a similar character identified in any tissue in the chronic study. Therefore, the relationship, if any, between belimumab treatment and the splenic abscesses identified in this isolated instance remains uncertain.

Microscopic Examination

Microscopic findings considered related to belimumab administration in the 4-week toxicology study included minimal to mild lymphoid depletion in B-lymphocyte areas of gut-associated lymphoid tissue (GALT) in the ileum and/or mesenteric lymph nodes but not in the spleen. These findings were noted in all belimumab-treated dose groups and, as illustrated in Table 3, were decreased in incidence after a 4-week treatment-free (recovery) period. These findings may represent an early response to belimumab treatment, as similar changes were not identified in the chronic administration study. In addition, treatment-related decreases in lymphocytes in the thymus, which primarily consists of T lymphocytes, were not recognized microscopically (Table 3).

At week 13 of the chronic study, microscopic findings attributed to belimumab administration consisted of decreased lymphoid follicle size and/or number in the spleen. A reduction in the size and/or number of lymphoid follicles in the spleen and mesenteric lymph node was also evident in belimumab-treated monkeys at week 26, as illustrated in Figure 8 and Table 3. This finding was not apparent after four weekly administrations of belimumab in the earlier study and, even after 26 weeks of belimumab treatment, was predominantly limited to the spleen. The density of lymphocytes and/or the

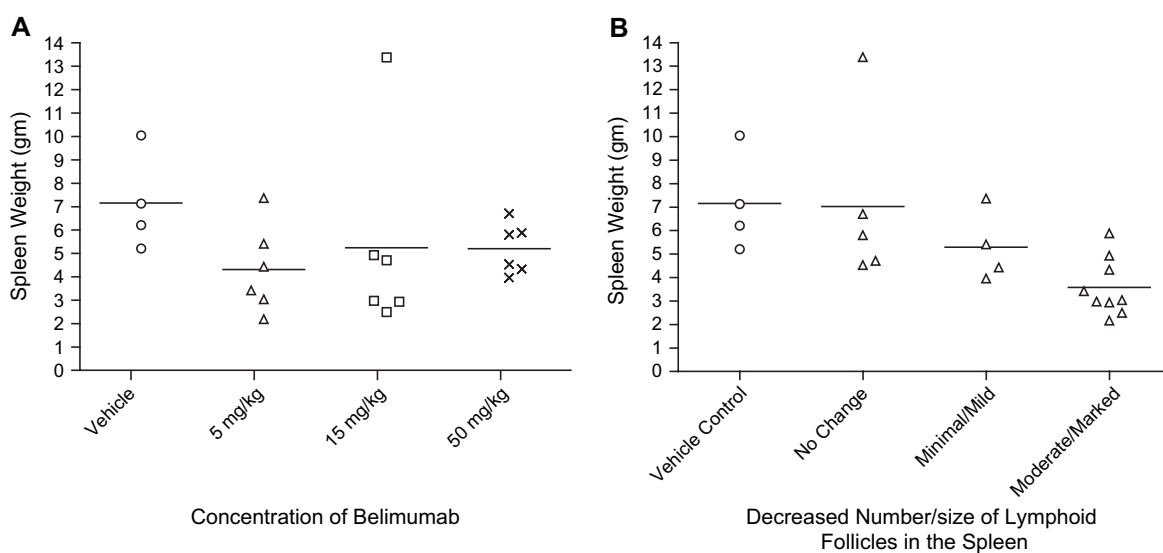


FIG. 7. In panel A, spleen weight is presented according to dose level for monkeys necropsied after 26 weeks of belimumab or vehicle control administration. Symbols: ○ = vehicle control; △ = 5 mg/kg/dose; □ = 15 mg/kg/dose; × = 50 mg/kg/dose. In panel B, the same data are presented according to microscopic evaluation of the spleen, rather than dose level, in belimumab-treated monkeys. Symbols: ○ = vehicle control; △ = belimumab treated. Data from the same 22 monkeys are presented in each panel; $n = 4$ for vehicle control, and $n = 6$ for each of the 5, 15, and 50 mg/kg dose levels of belimumab.

TABLE 3
Incidence of Minimal to Marked Decreases in Lymphoid Tissue Components in Cynomolgus Monkeys
According to Tissue, Dose and Schedule of Belimumab Treatment

Tissue	Dose level	Treatment schedule				
		4-week toxicology study ^a		13/26-week toxicology study		
		Q7d × 4	Q7d × 4 + 4 weeks	Q14d × 7 (week 13)	Q14d × 13 (week 26)	Q14d × 13 + 34 weeks (week 60)
Spleen	0 mg/kg	0 of 6	0 of 4	0 of 4	0 of 4	0 of 4
	5 mg/kg	0 of 6	0 of 4	2 of 6	6 of 6	0 of 4
	15 mg/kg	0 of 6	0 of 4	3 of 6	4 of 6	1 of 4
	50 mg/kg	0 of 6	0 of 4	1 of 6	3 of 6	0 of 4
	Total	0 of 24	0 of 16	6 of 22	13 of 22	1 of 16
Mesenteric lymph node	0 mg/kg	0 of 6	0 of 4	0 of 4	0 of 4	0 of 4
	5 mg/kg	1 of 6	1 of 4	0 of 6	1 of 6	0 of 4
	15 mg/kg	2 of 6	0 of 4	0 of 6	2 of 6	0 of 4
	50 mg/kg	5 of 6	1 of 4	0 of 6	2 of 6	0 of 4
	Total	8 of 24	2 of 16	0 of 22	5 of 22	0 of 16
Ileum (GALT)	0 mg/kg	0 of 6	0 of 4	0 of 4	0 of 4	0 of 4
	5 mg/kg	1 of 6	1 of 4	0 of 6	0 of 6	0 of 4
	15 mg/kg	3 of 6	0 of 4	0 of 6	0 of 6	0 of 4
	50 mg/kg	3 of 6	2 of 4	0 of 6	0 of 6	0 of 4
	Total	7 of 24	3 of 16	0 of 22	0 of 22	0 of 16
Thymus	0 mg/kg	4 of 6	3 of 4	0 of 4	2 of 4	0 of 4
	5 mg/kg	3 of 6	0 of 4	2 of 6	0 of 6	2 of 4
	15 mg/kg	3 of 6	1 of 4	2 of 6	0 of 6	1 of 4
	50 mg/kg	5 of 6	0 of 4	1 of 6	1 of 6	1 of 4
	Total	15 of 24	4 of 16	5 of 22	3 of 22	5 of 16

^aStudy design previously described (Baker *et al.*, 2003).

follicular size/number in other lymphoid organs examined histologically (thymus, mesenteric and mandibular lymph nodes, GALT) morphologically normal in most belimumab-treated monkeys, despite the decreased B-lymphocyte representation was determined by flow cytometry for mesenteric lymph node at week 26. The histomorphology of the spleen correlated with decreases in B-lymphocyte representation in spleen as determined by flow cytometry after 13 or 26 weeks of treatment with belimumab and, as previously mentioned, was correlated with reduced spleen weights after 26 weeks of belimumab treatment (Fig. 7B). Of monkeys treated with belimumab, the incidence of the splenic B-lymphocyte reduction was greatly reduced in monkeys necropsied in week 60, after a 34-week treatment-free recovery period; only one mid-dose (15 mg/kg/dose) monkey had a moderate decrease in lymphoid follicle size/number in the spleen after the 34-week treatment-free period. This isolated finding in this monkey may represent normal lymphoid tissue cycling, since other lymphoid tissues appeared normal in this monkey, and the percent CD20+ and CD20+/CD21+ lymphocytes had returned to baseline levels in peripheral blood and were similar to those of control monkeys in the spleen and mesenteric lymph node by flow cytometry (individual data not shown).

Serum Antibodies Specific for Belimumab

With chronic administration, antibodies specific for belimumab were detected in two of 60 monkeys during the 26-week treatment period. Positive monkeys included a male from the 5-mg/kg dose group and a female from the 50-mg/kg dose group. The appearance of anti-belimumab antibodies in these two monkeys coincided with decreased belimumab detected in serum, and a variable response to treatment. The low-dose (5 mg/kg) monkey's B-lymphocyte counts were considered essentially unaffected by belimumab, while the high-dose (50 mg/kg) monkey had a decrease in B lymphocytes similar to other high-dose monkeys treated with belimumab. All 16 monkeys evaluated at the end of the recovery period in this study were negative for belimumab-specific antibodies, including the two monkeys that were previously positive.

In addition, one monkey in a single-dose pharmacokinetic study developed antibodies to belimumab after a single iv injection at 50 mg/kg and had an altered pharmacokinetic profile. Therefore, three out of the 59 monkeys (5.1%) that were exposed to belimumab in these two studies developed a measurable antibody response that was specific for belimumab. None of the 12 control monkeys in these two studies developed antibodies to belimumab. In the 4-week study,

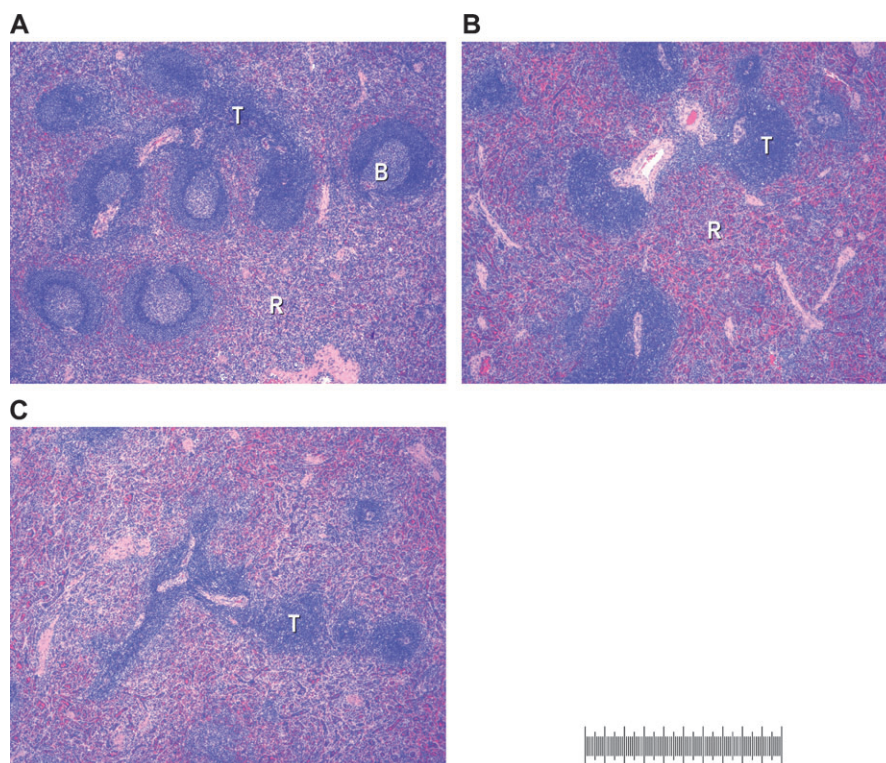


FIG. 8. Representative photomicrographs of monkey spleen (hematoxylin and eosin); 1-mm size bar photographed at the same magnification is included at the lower right. (A) illustrates typical spleen histomorphology in cynomolgus monkeys. A white pulp follicle, usually predominantly B lymphocytes, is indicated by a "B." A white pulp periaarteriolar lymphoid sheath, usually predominantly T lymphocytes, is indicated by a "T." The red pulp area is indicated by an "R." (B) and (C) illustrate spleens from monkeys treated with belimumab for 26 weeks at 5 or 50 mg/kg, respectively. Note relative absence of "B" areas in the spleens from monkeys treated with belimumab. Bar = 1 mm.

antibodies recognizing belimumab were identified in three of 30 monkeys exposed to belimumab but were not confirmed to be specific for belimumab and did not result in a concomitant alteration in belimumab PK.

Clinical Pathology

There were no serum chemistry, hematology, coagulation, or urinalysis findings that were considered to be related to belimumab administration (data not shown). There were slight to moderate increases in lactate dehydrogenase, aspartate aminotransferase, and/or alanine aminotransferase in individual monkeys in all groups, including controls, at one or more evaluation intervals throughout the chronic toxicology study. These sporadically increased serum enzyme activities are not uncommon in cynomolgus monkeys (Landi *et al.*, 1990) and were not considered related to treatment in these studies. In addition, increased white blood counts were noted in individual monkeys in all groups, including controls, at one or more evaluation time points, including prestudy. Changes in erythrocyte morphology (slight anisocytosis, slight poikilocytosis, hypochromasia) and platelet morphology (large platelets, platelet clumping) were noted occasionally during both toxicology studies; however, these changes occurred in

all groups (including the control group), and in the pre-study period, and are not considered related to belimumab administration.

Immunoglobulins

Immunoglobulin subclasses were analyzed as an end point of the 26-week toxicology study. The mean levels of each immunoglobulin subclass are presented in Figures 9A–D. The repeated measures model demonstrated a significant treatment effect for IgA (p value = 0.0065) and for total immunoglobulin (IgM + IgG + IgA; p value = 0.0391). However, there were no significant differences in IgA between belimumab-treated and control monkeys for any week during the study. For total immunoglobulin, there was evidence of treatment differences only at week 26. *Post hoc* tests indicated that 0 mg/kg (mean = 1553 mg/dl) was significantly greater than both the 5-mg/kg (mean = 1188 mg/dl, p value = 0.0250) and the 50-mg/kg (mean = 1231 mg/dl, p value = 0.0226) dose groups. However, the range of values for total immunoglobulin prestudy (week 1, 811–2312 mg/dl) and after initiation of treatment (weeks 4–60, 671–2562 mg/dl) was similar. Thus, this finding was considered of questionable biological significance.

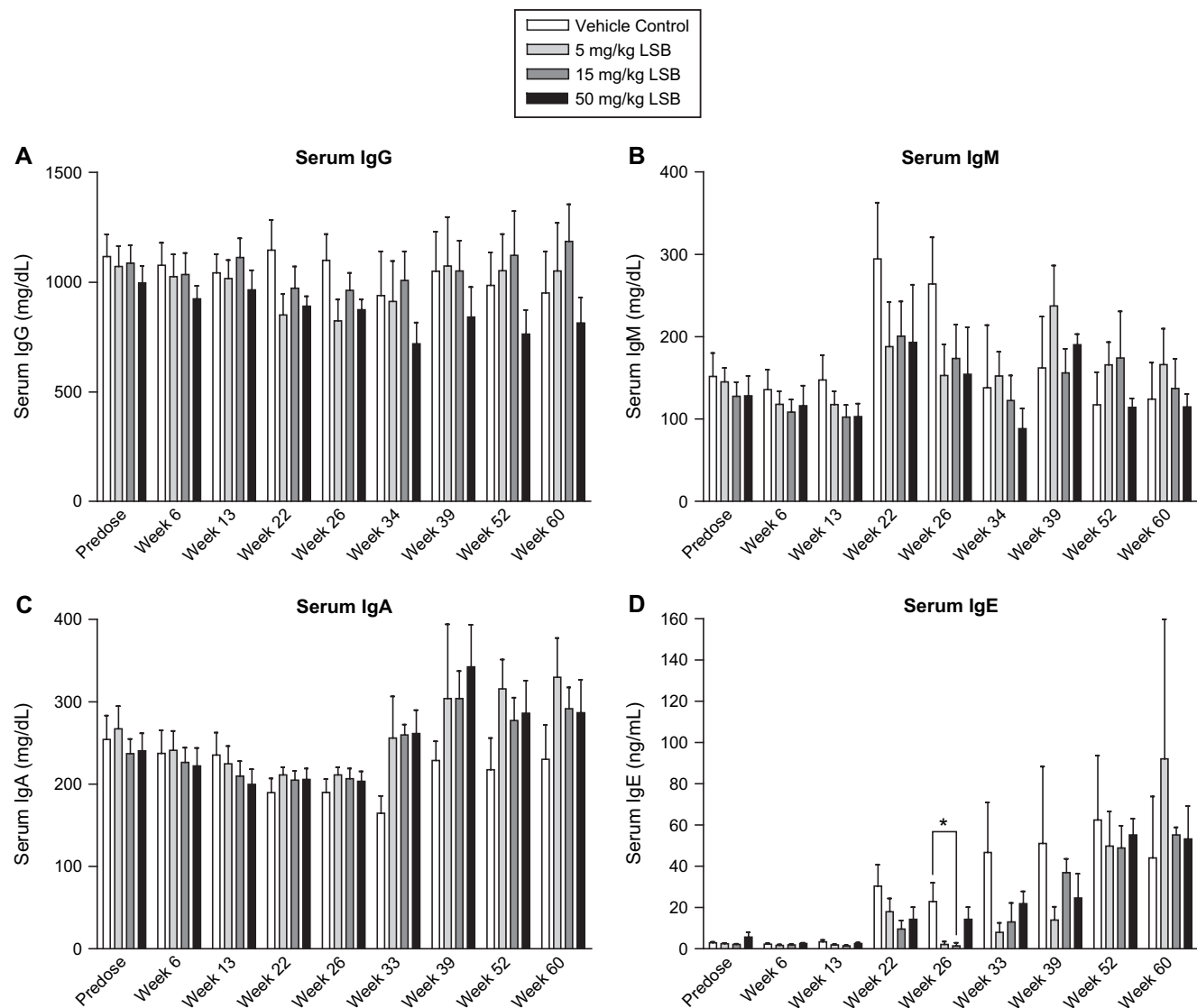


FIG. 9. The mean (+ SEM) value for each immunoglobulin class is indicated according to treatment group at each time point of assessment in the chronic toxicology study as follows: (A) Immunoglobulin G (mg/dl), (B) Immunoglobulin M (mg/dl), (C) Immunoglobulin A (mg/dl), (D) Immunoglobulin E (ng/ml). $n = 16$ monkeys per group through week 13, $n = 10$ monkeys per group from weeks 14 through 26, and $n = 4$ monkeys per group from weeks 27 through 60. Asterisks indicate significant differences in immunoglobulin subclasses in belimumab-treated monkeys as compared to vehicle controls ($p < 0.05$).

There were no statistically significant effects of belimumab on serum IgM or IgG in this study. Significant differences were observed in IgE only at week 26 and only between 0-mg/kg (mean = 23.1 ng/ml) and 15-mg/kg (mean = 1.6 ng/ml) dose groups (p value = 0.0496). There was a trend toward decreased IgE in the 5-mg/kg dose group as well but not for the 50-mg/kg dose group. Due to the high variability between IgE levels between monkeys, this finding was also considered to be of questionable biologic significance. An increase in IgE was noted across treatment groups, including the control group, and was therefore not considered related to belimumab administration.

DISCUSSION AND CONCLUSIONS

With administration for at least 13 weeks, belimumab specifically decreases lymphoid tissue and circulating peripheral blood B lymphocytes in cynomolgus monkeys. The effect on both tissue and peripheral blood lymphocytes was relatively consistent despite differences in absolute B-lymphocyte numbers between individual monkeys. The differences in B-lymphocyte representation between control and belimumab-treated monkeys were most prominent for the mature B-lymphocyte phenotype (CD20+/CD21+), suggesting that this is the most susceptible target for effects of BLyS depletion. Importantly, the total

lymphocyte count was not affected by belimumab, reflecting the limited and targeted activity of this antibody. Finally, although there were notable differences in belimumab effects between individual monkeys, there was no consistent dose-related difference in effect or in magnitude of the B-lymphocyte decrease over the 5- to 50-mg/kg dose range explored. The lack of belimumab dose relationship to B-lymphocyte decrease may indicate that these levels may be saturating for the target over this range and schedule in monkeys and also highlights the potential for variability in responses between individual monkeys at a given exposure.

The timing of the response, with a delayed onset in measurable effects and a protracted recovery period, is consistent with the hypothesis that BLyS antagonism indirectly decreases B lymphocytes, possibly through decreased survival and/or proliferation. The previously reported 4-week study demonstrated decreased B-lymphocyte representation in spleen and mesenteric lymph node mononuclear cells that were most apparent after 8 weeks (4 weeks of treatment followed by a 4-week treatment-free period), but no significant difference in peripheral blood lymphocytes. However, in the chronic study described here, both tissue and peripheral blood B lymphocytes were decreased after 13 weeks of exposure, and this finding was sustained through at least week 39, which was well into the recovery period.

The distribution of microscopic findings within the spleen after 26 weeks of belimumab exposure further supports the hypothesis that belimumab activity is specific for B lymphocytes. Interestingly, there was no apparent dose-response relationship to the severity of the decreases in lymphoid follicle size and number over the 5- to 50-mg/kg dose range evaluated in the chronic study. In addition, when considered by treatment group, any effect of belimumab on spleen weight was difficult to appreciate, in part due to the marked individual variability of this parameter in monkeys. However, when the spleen to body weight ratio was evaluated against the severity of the decreases in splenic follicular size and number rather than the dose level, an association of belimumab administration with decreased spleen weight becomes apparent. This effect was variable between individual monkeys treated with belimumab for 13 or 26 weeks and was not dose related over the 5- to 50-mg/kg dose range in this study. With one exception, spleens from monkeys treated with LymphoStat-B for 26 weeks, followed by a 34-week treatment-free (recovery) period, were histomorphologically similar to spleens from control monkeys. These data support the role of belimumab in the histomorphologic differences recognized at 26 weeks and suggest that individual animal sensitivity to belimumab may play a role in the effects of belimumab observed in cynomolgus monkeys.

Microscopically, changes in lymphoid tissue morphology were identified in GALT and mesenteric lymph node after four iv administrations as noted in Table 2 but appeared to be limited to the spleen after chronic administration of belimumab. There were decreases in the number and size of lymphoid

follicles in the white pulp of the spleen in many, but not all, of the monkeys treated with belimumab for 13 or 26 weeks, without a clear dose-response relationship to incidence or severity. However, there were no apparent histomorphologic differences across treatment groups in the other lymphoid tissues evaluated, such as lymph nodes, thymus, and GALT, at either 13 or 26 weeks. Thus, there may be differences in the principle lymphoid organs affected (GALT vs. spleen) depending on the length of treatment with belimumab. Such anatomic differences in B-lymphocyte sensitivity have also been identified with anti-CD20 antibodies and may reflect differential dependence of these populations on survival factors such as BLyS (Gong *et al.*, 2005).

There were no clear effects of belimumab on basal immunoglobulin levels in cynomolgus monkeys in either the 4-week or the 26-week toxicology study. Serum immunoglobulins tended to fluctuate during the course of the chronic toxicology study. In general, there were no findings clearly attributable to belimumab treatment, although there was a general trend for IgM and IgG to be decreased in belimumab-treated monkeys at 22 and 26 weeks and for IgA to be increased in belimumab-treated monkeys during the recovery period. Isolated differences for different groups versus control monkeys were identified, but without a temporal or dose-dependent pattern indicating a specific effect of belimumab. The lack of effect of belimumab on basal immunoglobulin is consistent with data from other antibody therapeutics targeting B lymphocytes (Gong *et al.*, 2005; Martin and Chan, 2004). Interestingly, IgE levels increased markedly over the 14-month period of the study, but this was noted for all treatment groups, including control monkeys. Therefore, this finding may reflect changes in environment, or possibly the immunologic maturational state, of the monkeys in this chronic study. Neither serum globulin nor the albumin:globulin ratio was affected by belimumab administration (data not shown).

Although the presence of infections in even a small number of monkeys in each study might suggest possible immunosuppression, a clear relationship between infection and belimumab administration could not be established. The multifocal abscesses identified in the spleen of one high-dose (50 mg/kg) monkey in the 4-week study were an isolated finding. This monkey had a mild peripheral neutrophilia for at least 2 weeks prior to belimumab administration and had consistently elevated serum IgG prior to and throughout the study, suggesting that the splenic abscessation may have been a preexisting condition in this monkey. In the 26-week study, there were cage side observations of red nasal discharge suggestive of *Branhamella* infection across all treatment groups, including controls. When antibiotic intervention was warranted for the affected monkeys, all monkeys appeared to respond appropriately to that therapeutic intervention regardless of treatment group. It should be noted that *Branhamella* has been causatively linked to the bloody nose syndrome cynomolgus macaques (VandeWoude and Luzarraga, 1991), is reported

sporadically in macaque colonies (reviewed in Olson and Palotay, 1983), and is occasionally observed at Charles River Laboratories Preclinical Services.

The pharmacokinetics profiles were linear across the 30-fold dose range evaluated in the single-dose studies. The volume of distribution is relatively small but exceeds the plasma volume, indicating that belimumab distributes to tissues. Belimumab total body clearance is substantially lower than the glomerular filtration rate for monkeys (2995 ml/day/kg), indicating little renal clearance of belimumab. The half-life is long, ranging between 9 and 16 days.

In the 26-week multiple-dose toxicology study detailed here, all monkeys in belimumab-treated groups appear to have been exposed to appropriate levels of drug, with the exception of the two anti-belimumab antibody positive monkeys. Exposure of belimumab increased with dose and extended into the recovery period. The amount of drug accumulation at steady state (approximately twofold) was consistent with the dosing regimen and expected half-life and PK appeared to be linear over the range of doses tested. The pharmacokinetic behavior after single- and multiple-dose administration is consistent with large macromolecules such as other monoclonal antibodies (Gobburu *et al.*, 1998).

In conclusion, belimumab was well tolerated by all monkeys and demonstrated the expected pharmacologic activity of specifically decreasing B lymphocytes both in tissues and in peripheral blood. Toxicity of belimumab was not identified with chronic administration. Exposures in these nonclinical monkey studies included levels 5- to 10-fold higher than exposures attained in phase 2 clinical studies. In addition, there was minimal immunogenicity of belimumab detected in this chronic study. Finally, the treatment-related effects of belimumab were reversible after a treatment-free recovery period.

ACKNOWLEDGMENTS

The authors would like to thank Yuling Li and Melissa Perkins for the production and formulation of the belimumab used in these studies. Todd Riccobene provided the initial pharmacokinetic analyses for the repeat-dose studies; Jean Recta provided the initial statistical analysis of immunoglobulin subclass levels. Greg Beattie and Nancy Gillett contributed to study execution and interpretation of findings in the toxicology studies. Youmei Wu performed ELISA measurements of affinities with human and cynomolgus BLyS proteins. The authors also thank Jim Fikes, Thi Sau Migone, Paul Moore, and David Hilbert for helpful discussions in the development of the manuscript and Terry Manspeaker for excellent assistance in preparing the figures.

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